



COURSE NUMBER: CP1410

COURSE TITLE: E-Commerce Web Analysis and Design

COURSE DESCRIPTION:

This course introduces students to the concepts of systems analysis and design for the Web. It gives a fundamental overview of the Web site development process, and details the iterative cycle of planning, analysis, design and development, and testing. Emphasis is placed on designing an effective, user-centered, accessible commerce based Web site.

PREREQUISITES: CR1511 - Website Development

CO-REQUISITES: None

CREDIT VALUE: Four (4)

COURSE HOURS PER WEEK: Three (3)

LAB HOURS PER WEEK: Two (2)

REQUIRED TEXT: None

LEARNING RESOURCES:

Lynch, P.J., and Horton, S. (2010). *Web style guide: Basic design principles for creating Web sites* (4th ed.). Yale University Press. Retrieved from <http://www.webstyleguide.com>

Siegel, Ben (February 20, 2018) Smashing Magazine: A Comprehensive Website Planning Guide <https://www.smashingmagazine.com/2018/02/comprehensive-website-planning-guide-part1/>

Stanford University Continuing Studies CS 22 Enhanced Web Site Design:
<https://web.stanford.edu/group/csp/cs22/pdfs/webdesign>

W3C Accessibility, Usability, and Inclusion
: <https://www.w3.org/WAI/fundamentals/accessibility-usability-inclusion/>

U.S. Department of Health and Human Services, Web Communications and New Media Division. (n.d.). *Usability.gov: Your Guide for Developing Usable and Useful Web Sites*. Retrieved from <http://www.usability.gov/methods/process.html>

MAJOR TOPICS:

- 1.0 Introduction to Web Analysis and Design
- 2.0 Web Site Planning
- 3.0 Web Site Analysis
- 4.0 Web Site Design and Development
- 5.0 Web Site Testing and Refinement

LEARNING OBJECTIVES:

1.0 Introduction to Web Analysis and Design

- 1.1 Describe the steps in the Web site development process
- 1.2 Differentiate models of design
 - 1.2.1 Describe the linear model of design
 - 1.2.2 Describe the iterative model of design
 - 1.2.3 Explain why the iterative model of design is best suited for Web site development projects
- 1.3 Discuss software methodologies in E-Commerce
- 1.4 Describe Agile methodologies
- 1.5 Describe Waterfall methodology
- 1.6 Discuss Change Management

2.0 Web Site Planning

- 2.1 Discuss Web Site design
 - 2.1.1 Describe technical issues such as bandwidth, browser incompatibilities, and screen resolution
 - 2.1.2 Describe the importance and use of Web standards, and the role of the World Wide Web Consortium (W3C)
 - 2.1.3 Describe accessibility and usability issues, and the role of the Web Content Accessibility Guidelines (WCAG) and Section 508 guidelines
 - 2.1.4 Describe legal and privacy issues such as copyright, intellectual property, content liability, and personally identifiable information data security
- 2.2 Plan an E-Commerce/Web project
 - 2.2.1 Conduct a needs assessment to determine client needs and Web site scope, using one or more data-gathering techniques, such as:
 - 2.2.1.1 Survey
 - 2.2.1.2 Interviews
 - 2.2.1.3 Focus groups
 - 2.2.2 Choose a name for the site
 - 2.2.3 Describe the process of acquiring a domain name
 - 2.2.4 Describe the purpose of the site

- 2.2.5 Identify the site's audience
- 2.2.6 Develop a prioritized list of goals and objectives for the site
- 2.2.7 Prepare a project budget and schedule
- 2.2.8 Describe the roles and responsibilities of the site development team

3.0 Web Site Analysis

- 3.1 Use one or more data-gathering techniques to collect information about the target audience
- 3.2 Conduct task analysis to understand what users want from the site
- 3.3 Create personas to describe representatives of your target audience
- 3.4 Create scenarios to describe common user tasks on the site
- 3.5 Create user archetypes
- 3.6 Create usability tests for each user archetype
- 3.7 Identify similar and competing sites
 - 3.7.1 Evaluate similar and competing sites
 - 3.7.2 Analyze the current site, if one exists
- 3.8 Identify site requirements based on user scenarios
 - 3.8.1 Identify necessary technical requirements, including:
 - 3.8.1.1 Databases
 - 3.8.1.2 Scripting
 - 3.8.1.3 Hosting requirements

4.0 Web Site Design and Development

- 4.1 Discuss information architecture requirements
 - 4.1.1 Develop a content inventory
 - 4.1.2 Define the terms:
 - 4.1.2.1 Metadata
 - 4.1.2.2 Hierarchy
 - 4.1.2.3 Taxonomy
 - 4.1.2.4 Controlled vocabulary
 - 4.1.3 Identify different types of site structures, including:
 - 4.1.3.1 Sequences
 - 4.1.3.2 Hierarchies
 - 4.1.3.3 Webs
 - 4.1.4 Conduct a card sort to organize and group site content
 - 4.1.5 Diagram the site structure using site maps and wireframes
 - 4.1.6 Create a storyboard demonstrating site flow
 - 4.1.7 Develop the site file and directory structure
- 4.2 Analyse effective interface designs
 - 4.2.1 Describe the different types of navigation including:

- 4.2.1.1 Global
 - 4.2.1.2 Local
 - 4.2.1.3 Network
- 4.2.2 Differentiate types of navigation systems including:
 - 4.2.2.1 Text and image links
 - 4.2.2.2 Tabs
 - 4.2.2.3 Menus,
 - 4.2.2.4 Bars
 - 4.2.2.5 Breadcrumb trails
 - 4.2.2.6 Site maps
 - 4.2.2.7 Searches
- 4.2.3 Discuss interface design best-practices and web conventions
 - 4.2.3.1 Apply interface design best-practices and web conventions
- 4.2.4 Apply best-practices for web-based form design
- 4.2.5 Create use cases to describe how users will perform tasks on the site
- 4.2.6 Discuss parallel design
- 4.3 Demonstrate basic visual design principles, including:
 - 4.3.1 Consistency
 - 4.3.2 Contrast
 - 4.3.3 Repetition
 - 4.3.4 Alignment
 - 4.3.5 Proximity
 - 4.3.6 White space
 - 4.3.7 Color
 - 4.3.8 Typography
 - 4.3.9 Fixed layouts
 - 4.3.10 Fluid layouts
- 4.4 Build a prototype of a site for testing purposes
 - 4.4.1 Employ design techniques to ensure the site is (but not limited to):
 - 4.4.1.1 Fast-loading
 - 4.4.1.2 Compatibility
 - 4.4.1.3 Reliability
 - 4.4.1.4 Accessibility
 - 4.4.1.5 Useable
 - 4.4.1.6 Compliant

5.0 Web Site Testing and Refinement

- 5.1 Describe the process of conducting usability testing
 - 5.1.1 Create a usability testing plan
 - 5.1.2 Conduct a usability test on the site design

5.1.3 Analyze and report the usability test results

5.2 Perform technical testing of the site on a variety of different browsers, platforms, displays, and bandwidths

EVALUATION:

Assignments	15%
Test(s)	25%
Term Project	30%
Final Exam	30%

DATE DEVELOPED: March 2010

DATE REVIEWED:

REVISION NUMBER: 2

DATE REVISED: October 2020

Effective PeopleSoft Date: September 2021

Note to instructor: Check PIRS to ensure this outline is the most current version.